

Product Requirement Document (PRD)

WhatsApp Smart Search AI

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1. Overview

WhatsApp has become the default platform for sharing resumes, PDFs, internship links, assignments, tickets, and important documents. Current search relies on exact keyword matching. This feature introduces AI-powered semantic search that understands natural language and retrieves messages, files, and media using context rather than exact keywords.

2. Problem Statement

Users remember who sent a file, what it was about, and roughly when it was shared, but rarely remember exact filenames or message text. This causes significant time spent manually scrolling chats.

3. Goal

Reduce the effort required to find important information by enabling context-aware semantic search.

4. Success Criteria

Users should retrieve the desired result within the first three search results using natural language queries.

5. Target Users

Primary: Students, Working Professionals. Secondary: Small Business Owners, Recruiters, Project Teams.

6. User Personas

Persona 1: Engineering student searching for resumes, professor messages and internship links.

Persona 2: Working professional searching HR documents, invoices and meeting files.

7. User Stories

As a student, I want to search 'Resume I sent to professor' so I don't need to remember filenames.

As a professional, I want AI to understand natural language. As a recruiter, I want important files ranked above casual chats.

8. Current User Flow

Open WhatsApp → Search → Type keyword → No results → Try another keyword → Scroll chats → Find document.

9. Proposed User Flow

Open WhatsApp → Smart Search → Type natural language query → AI understands intent → Ranks relevant results → Shows documents, messages, links and images → User opens result.

10. MVP Features

Natural Language Search, Semantic Ranking, Document Search, Image Search (metadata), Conversation Summary, Smart Filters.

11. Out of Scope

Voice Search, OCR inside images, AI Chatbot, Cross-device semantic sync, AI-generated replies.

12. Functional Requirements

Understand natural language, retrieve messages/documents, rank by relevance, show previews, display context, respond quickly.

13. Non-functional Requirements

Search latency under 2 seconds, privacy-preserving, scalable, minimal battery impact.

14. Edge Cases

Deleted messages, duplicate filenames, multiple similar PDFs, offline mode, empty search, incorrect prediction, large groups, very old conversations.

15. Assumptions

Users remember context better than filenames. Relevant results are preferred over many weak matches. Privacy should remain unchanged.

16. Risks

Incorrect ranking, false positives, battery usage, privacy concerns, processing cost.

17. Success Metrics

North Star: Search Success Rate. Supporting: Average Search Time, Daily Smart Searches, Repeat Usage, Successful Result Click Rate. Guardrails: Latency, Battery Usage, Wrong Result Rate, Privacy Complaints.

18. Rollout Strategy

Phase 1: Internal Prototype → Phase 2: Student Beta → Phase 3: Power Users → Phase 4: Global Rollout.

19. Future Scope

Voice Search, OCR Search, Image Understanding, Calendar Integration, Cross-app Search, AI Recommendations.

20. Open Questions

Should Smart Search be enabled by default? Should AI process everything locally? How should deleted messages be handled? Should relevance outweigh recency?